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The relation between young adults' trust beliefs in others and interpersonal hostility

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Abstract

The research examined whether, and if so how, young adults' trust beliefs in others were associated with interpersonal hostility. The participants in Study 1 were 139 young adults from the UK (76 women; Mage = 20.8). In Study 2, 88 young adult women from the UK (Mage = 21.5) served as participants. The participants completed a standardized measure of trust beliefs in others (total with reliability, honesty, and emotional subscales). In Study 1, participants imagined they were victims of peer provocation. They were required to judge the intention for the provocation and their retaliation to it. In Study 2, the participants were engaged in a lab-based acquaintanceship interaction that involved the exchange of disclosures. They completed an adjective checklist that assessed anger and evaluated the quality of the conversation. Trust beliefs were linearly and negatively associated with the attribution of hostile intentions, retaliation, anger toward others, and critical evaluation of a developing peer relationship. As expected though, quadratic relations were found. Young adults with very low and those with very high trust beliefs (primarily emotionally based) showed greater attribution of hostile intentions, retaliation, anger toward others, and critical evaluation of a developing peer relationship than did young adults with the middle range of trust beliefs. The linear relations supported the hypothesis that trust promotes psychosocial adjustment. The quadratic relations supported the deviation from the normative trust (centralist) approach primarily for emotional trust beliefs in others.

KEYWORDS

anger, attribution of hostile intentions, critical evaluation of peer relationships, retaliation, trust beliefs, young adults

The principle that trust promotes psychosocial adjustment (TPPA) is the cornerstone of Attachment Theory (Cohn, 1990; De Winter et al., 2016), Generalized Expectancy Theory (Rotter, 1980), and Social Capital Theory (Cozzolino, 2011). As support for the TPPA, individuals' trust beliefs in others have been found to be concurrently and prospectively associated with prosocial behavior (Malti et al., 2016), lower levels of internalized psychopathology, and lower levels of externalized psychopathology (Malti et al., 2013; Rotenberg et al., 2005). The contrasting Centralist Approach to Trust (CAT) conceptualizes trust as a continuum in which deviation from normative patterns of trust beliefs and behavior results in psychosocial maladjustment (Rotenberg, unpublished). CAT is supported by quadratic relations in which children and early adolescents with very high trust and those with very low beliefs in others demonstrate greater psychosocial maladjustment (e.g., retaliatory aggression) than those with the middle range of trust beliefs in others (Rotenberg et al., 2013).

The current research was designed to further investigate the contrasting TPPA and CAT hypotheses. The research examined whether there were linear and curvilinear relations between trust

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beliefs in others and interpersonal hostility in young adults. Interpersonal hostility was assessed by the attribution of hostile intentions to others, (reported) retaliation, anger, and critical evaluation of a developing peer relationship. The research examined specific ways that those aspects of interpersonal hostility serve as mediators of the relations with trust beliefs in others. Finally, the study examined the role of different bases of trust beliefs in the relations and the implications of social histories for the relations in young women. The research utilized both a vignette paradigm and a lab-based method of relationship development.

1 | CONCEPTUAL FRAMEWORK FOR THE INVESTIGATION OF TRUST BELIEFS IN OTHERS

Trust beliefs in others were conceptualized by the Basis, Domain, and Target Interpersonal Trust Framework (BDT; Rotenberg, 2010, 2019). According to the BDT framework, trust beliefs comprise individuals' expectations that other persons show three classes (bases) of behavior: reliability, emotional, and honesty. The reliability basis of trust beliefs comprises individuals' expectations that other persons show reliability by fulfilling their word or promise. The emotional basis of trust beliefs comprises individuals' expectations that other persons refrain from emotional harm including refraining from causing emotional embarrassment, uncritically accepting personal disclosures, and maintaining their confidentiality. The honesty basis of trust beliefs comprises individuals' expectations that other persons tell the truth as opposed to lying and engage in behavior guided by benign rather than malevolent and sincere rather than insincere motivations. According to the BDT framework, individuals' trust beliefs in other persons vary by the target dimensions of familiarity and specificity. The current research investigated the implications of the three bases of trust beliefs and their total in others by young adults for their interpersonal hostility.

2 | TPPA AND RELATIONS BETWEEN TRUST BELIEFS AND INTERPERSONAL HOSTILITY

Retaliatory aggression (including reactive aggression) is harmful behavior that typically involves the emotional state of anger (Raine et al., 2006; Wilkowski & Robinson, 2010). Meta-analyses have shown that retaliatory aggression is concurrently and prospectively associated with psychosocial maladjustment, internalizing psychopathology, and peer rejection (see Card & Little, 2006). Males demonstrate more direct aggression and retaliatory aggression than do females during childhood (Bondü & Richter, 2016; Card et al., 2008; Kempes et al., 2006). Men demonstrate greater direct aggression than women but under specific conditions. The sex differences are evident primarily when the individual was unprovoked (Bettencourt & Miller, 1996) and when the aggression caused pain rather than psychological or social harm (Eagly & Steffen, 1986). A comprehensive meta-analysis by Archer (2019) confirmed that men showed greater (1) verbal aggression (small effect size), (2) physical aggression (medium effect size), weapon use (very large effect size), (c) violent crime (very large effect size), (d) violent computer game use (very large effect size), (e) revenge (large effect size) and (f) a social dominance orientation (medium effect size).

The relation between children's trust beliefs in others and retaliatory aggression has been examined. The Hostile Attribution Bias (Dodge, 1980, 2006; Orobio de Castro et al., 2002) is a distortion-cue process in which aggressive individuals attribute hostile intentions to ambiguously motivated provocation and retaliate to it. Dodge (1980) found that Hostile Attribution Bias in aggressive children was linked to their mistrust of the intentions of peer perpetrators. Research has demonstrated that children who hold low trust beliefs in peers (primarily reliability-based) showed stable and increasing developmental trajectories of aggression (Malti et al., 2013). Malti et al. (2013) proposed that trust beliefs in others reduced aggression because it decreased individuals' attribution of intention to the provocation. Research has demonstrated that children's mistrust in others is associated with externalizing problems comprising aggression and callous-unemotional traits (Wong et al., 2014).

The relation between adults' trust beliefs in others and their aggression and anger has been examined but in a limited fashion. Young adults' perceptions of mistrust/abuse in others have been found to be associated with trait measures of aggression (Tremblay & Dozois, 2009) and acceptance of aggression (Cadenhead & Richman, 1996). Berends et al. (2019) found that a history of aggressive behavior in healthy young men was associated with a combination of low trait measures of trust and low levels of oxytocin. Adults' mistrust in the opposite sex is associated with being a victim of intimate partner aggression (Copp et al., 2017; LaMotte et al., 2016).

The aforementioned research supports the TPPA hypothesis by showing that trust beliefs in others are negatively associated with interpersonal hostility. There are, however, four gaps in knowledge that were redressed by the current research. First, it examined whether young adults' trust beliefs in others are negatively associated with retaliatory aggression rather than broader trait-like measures of aggression. Second, guided by the study by Malti et al. (2013), the research examined the hypothesis that young adults' trust beliefs in others reduced the likelihood of retaliatory aggression because those beliefs reduced the tendency for them to attribute hostile intentions to provocation. Third, the current research examined whether relations between trust beliefs in others and retaliatory aggression are the results of a cue-distortion process as implied by Hostile Attribution Bias rather than a comprehensive personal style. Trust beliefs in others should be negatively associated with attribution of intention and retaliatory aggression only for ambiguously motivated provocation according to a cue-distortion process. If the relations are due to a comprehensive personal style, then trust beliefs in others would be negatively associated with attribution of intention and retaliatory aggression to hostilely motivated provocations as well. Fourth, if trust beliefs in others are

negatively associated with interpersonal hostility then that relation could generalize to anger toward others and critically evaluating a developing relationship.

3 | CAT AND THE RELATION BETWEEN TRUST BELIEFS AND INTERPERSONAL HOSTILITY

According to CAT, extreme levels of trust beliefs and behavior detract from normative patterns of social relationship functioning resulting in poor psychosocial adjustment (Rotenberg, 2021). The CAT specifies the processes responsible for the poor psychosocial adjustment shown by individuals at the two ends of a continuum of trust beliefs in others. At the lower quadrant of trust belief continuum (i.e., very low trust beliefs), individuals hold paranoid-like mistrusting cognitions and behavior that disposes them to attribute hostile intentions to provocation, engage in retaliatory aggression, experience anger toward others, and withdraw from social relationships (see Bebbington et al., 2013; Edens et al., 2009; Hallquist & Pilkonis, 2012; Wong et al., 2014). As a result of that predisposition, those individuals are not inclined to develop and maintain satisfactory social relationships. At the upper quadrant of the trust belief continuum (i.e., very high trust beliefs), individuals experience incongruence between their trust beliefs in others and the normative level of trustworthiness in society. As a consequence, they frequently experience betrayal and disappointment which causes them to establish an interpersonal hostility style that includes anger towards others and hypercritical evaluations of social interactions. As a consequence, they are not inclined to develop and maintain satisfactory social relationships. According to CAT, the effects of trust beliefs in others on poor psychosocial adjustment are compounded by the social stereotyping of persons at two ends of the continuum of trust beliefs in others. Individuals with very low trust beliefs are often viewed by others as cynics and thus tend to be socially rejected. Also, individuals with very high trust beliefs are often viewed by others as suckers and experience social rejection.

In support of CAT, curvilinear (quadratic) relations have been found between trust beliefs in others and aggression during childhood and early adolescence. Children and adolescents with very high trust beliefs and those with very low trust beliefs showed greater internalized maladjustment across time, rejection by peers, and exclusion by peers than did children and early adolescents with the middle range of trust beliefs, (Rotenberg et al., 2010). Children with very high and those with very low trust beliefs in peers showed greater direct aggression during school recesses than did children with the middle range of trust beliefs (Rotenberg et al., 2014). Early adolescents with high and those with very low trust beliefs in peers reported greater retaliatory aggression to vignette peer provocation than did those with the middle range of trust beliefs (Rotenberg et al., 2013).

The CAT and related research led to the currently proposed hypotheses regarding the relation between trust beliefs in others and interpersonal hostility in young adults. Young adults with very low and those with very high trust beliefs in others would be inclined to show greater interpersonal hostility on the measures than would young adults with a middle range of trust beliefs. It was anticipated that the quadratic relations would be found primarily with emotional trust beliefs in others notably as the result of the betrayal experienced by young adults with very high trust beliefs in others. For example, it was anticipated that young adults with very high trust beliefs would perceive aggression acts by peers as betrayal (i.e., "I was harmed by peers despite my emotional trust in them").

4 | WOMEN, SOCIAL HISTORY, AND QUADRATIC RELATIONS

This current research examined the relations between trust beliefs and both anger toward others and critical evaluation of a developing peer relationship in women. The single-gender investigation was guided by the finding that during childhood, girls show a specific set of quadratic relations between trust beliefs in others (notably peers) and their peer interactions (Rotenberg et al., 2014). It was found that girls with very high trust beliefs and those with very low beliefs engaged in greater indirect aggression, elicited lower group affiliation, incurred greater peer rejection, and showed greater distress in peer interactions than did girls with the middle range of trust beliefs. The findings are consistent with CAT. Specifically, girls with low trust beliefs in others engaged in hostile peer interactions (including indirect aggression) which elicited social exclusion. Also, girls with high trust beliefs in others were similarly involved in indirect aggression which caused them to experience distress. There is evidence that trust beliefs are modestly stable across time (Rotenberg et al., 2010), and therefore, the social experiences of girls could be repeated across the course of development. The current research was designed to examine the developmental consequences of these childhood experiences for young adulthood. It was expected that the women would establish anger towards others and critical evaluations of developing which would be shown in an acquaintanceship lab-based method particularly suited for university women. The acquaintanceship method entailed the participant's report of her emotions before the development of a peer relationship and the evaluation of that relationship. Research shows that young women frequently exchange self-disclosures with same-sex peers in the development of relationships (Dindia & Allen, 1992). Women in university are often asked to exchange disclosures to develop peer relationships in first-year receptions.

5 | OVERVIEW OF THE CURRENT RESEARCH AND HYPOTHESES

Young adults completed a standardized measure of trust beliefs in others (total, as well as reliability, emotional, and honesty subscales). In Study 1, participants imagined they were victims of peer provocation. They were required to judge the intention for the provocation and their retaliation to it. In Study 2, women were engaged in the lab-based participant. They reported their emotional states (including anger), exchanged disclosures

with a peer as part of relationship development, and evaluated the quality of the conversational exchange.

The following were predicted guided by the two theoretical approaches:

5.1 | TRUST PROMOTES PSYCHOSOCIAL ADJUSTMENT (TPPA) HYPOTHESES

(a) Trust beliefs would be negatively associated with the attribution of hostile intentions for provocation and retaliation to it. The attribution of hostile intentions would mediate the relation between trust beliefs in others and retaliation (Hypothesis 1: trust-intention-retaliation).
(b) Trust beliefs would be negatively associated with anger and critical evaluation of a peer relationship. Anger would mediate the relation between trust beliefs in others and critical evaluation (Hypothesis 2: trust-anger-critical evaluation).

The research explored the implications of the Hostile Attribution Bias as compared to the comprehensive personal style regarding the relation between trust beliefs in others and both the attribution of hostile intentions and retaliation.

5.2 | CENTRALIST APPROACH TO TRUST (CAT) HYPOTHESES

(a) Quadratic relations would be found between trust beliefs and both the attribution of hostile intention for provocation and retaliation. Young adults with very high trust beliefs and those with very low trust beliefs would show higher attributions of intention for provocation and retaliation than would young adults with the middle range trust beliefs in others (Hypothesis 3: trust-retaliation quadratic).
(b) Quadratic relations would be found between trust beliefs and both anger and critical evaluation in women. Women with very high trust beliefs and those with very low trust beliefs would show higher anger and critical evaluation of peer relationship development than would women with the middle range trust beliefs (Hypothesis 4: trust-anger-critical quadratic).
(c) The preceding quadratic relations would be found primarily with emotional trust beliefs in others (Hypothesis 5: emotional betrayal).

6 | STUDY 1

6.1 | Method

6.1.1 | Participants

One hundred and thirty-nine young adults (76 women, $M = 20.8$ years, $SD = 5.5$) enrolled in a mid-sized university in the UK were participants. They were solicited by advertisements on a university-wide sited. This was an opportunity sample. The research was approved by the appropriate institutional ethics committees. APA and BPS ethics principles of research conduct were adhered to.

6.2 | Measures

6.2.1 | Trust beliefs in others

The generalized trust beliefs scale – late adolescence (GTBS – LA; Randall et al., 2010) assesses three BDT bases of trust beliefs (reliability, emotional, and honesty, as well as total) in four significant others (mother, father, romantic partner, and peers). The GTBS – LA has shown acceptable internal consistency ($\alpha > .80$) and structure (Randall et al., 2010). The items are judged on a 5-point Likert scale (very unlikely to very likely). Examples of the items are (with basis and target in brackets): (1) "Luke's mother told Luke she would pay for all of his textbooks for the next semester if he helped clean out their garage for the winter. Luke helped his mom clean out their garage. How likely is it that Luke's mother will pay for all of his textbooks? (reliability-mother), and (2) "Parker tells his girlfriend, Rhonda, that he failed general biology and asks her not to tell anyone else. How likely is it that Rhonda will not tell anyone else?" (emotional-romantic partner).

A confirmatory factor analysis using structural equation modeling (SEM) was performed on the GTBS-LA items. The predicted model was composed of three latent factors (reliability, emotional, and honesty) with paths from each of the four targets (mother, father, peer, and romantic partner) to each factor. The model included covariances between each of the three latent factors and error terms (Byrne, 2001). The SEM analysis yielded, $\chi 2$ (46) = 71.18, $p = .01$, normed fit index (NFI) = 0.87, comparative fit index (CFI) = 0.92, and root mean square error of approximation (RMSEA) = 0.063. All the paths between the raw items and their intended latent factors attained significance at $p < .05$ as did the covariances between the three latent factors. SEM analyses also showed that two-factor and one-factor models of the scale did not fit the data according to any of the criteria. The expected three-factor model was an adequate account of the data because it met the criteria, CFI > 0.90, RMSEA < 0.07, and other simpler models did not fit the data (Byrne, 2001; Hu & Bentler, 1999). The chi-square was statistically significant (contrary to fitness criteria) but that is frequently found when sample sizes exceed 100, as was the case in the current study. The total GTBS-LA scale demonstrated acceptable internal consistency $\alpha = .78$. Each of the three trust belief subscales also demonstrated acceptable internal consistency given the limited number of items per subscale, $\alpha s = 60$, .64 and .64. Larger scores on scale/subscales denoted greater trust beliefs.

6.2.2 | The vignettes

The vignettes used with early adolescents by Rotenberg et al. (2013) were modified for young adults. Six hypothetical vignettes were presented that depicted three types of provocation by peers (physical,

verbal, and social exclusion). The victims and the perpetrators were identified as the same-sex as the participants. The vignettes depicted two types of motivation of the perpetrator (hostilely motivated and ambiguously motivated). Examples of the vignettes for men (with the type of provocation identified in brackets) are "Whilst the students were eating in the lunchroom, Chris spilt his coffee over Luke" (ambiguously motivated), and "Stuart is hosting an end of year party; he invites everybody he knows except for Phil" (hostilely-motivated). Participants were asked to imagine that they were the victim in each vignette and to judge the vignette on five questions using a 5-point Likert scale (1 – very likely to 5 – very unlikely).

6.2.3 | Attribution of (hostile) intention

Three questions assessed attributions of intention: (1) "To what extent do you think that the perpetrator (vignette name) meant to harm the victim either physically, verbally, or emotionally?", (2) "To what extent do you think the perpetrator (vignette name) did this to you on purpose?", and (3) "To what extent do you think that the perpetrator's (vignette name) actions are hostilely motivated? The ratings for the three questions were summed. Higher scores denoted greater reported attribution of intention. The next set of questions assessed retaliation.

6.2.4 | Retaliation (retaliatory aggression)

Two questions assessed retaliation: (1) "If you were the victim (vignette name), how likely would it be for you to try and get back at the perpetrator?", (2) "if you were the victim (vignette name), how likely would it be for you to attempt to harm the perpetrator either physically or emotionally? Emotional means hurting someone's feelings". The ratings for the two questions were summed. Higher scores denoted greater retaliation.

6.3 | Procedure

The participants completed the scales and reported their age and gender on surveys accessed online on a university-wide intranet network site. This method provides the advantage of testing a diverse and comprehensive sample while ensuring confidentiality (Evans & Mathur, 2018). The study was described as an investigation of peoples' attitudes and behavior toward others.

6.4 | Results and discussion

6.4.1 | Strategies for the analyses

First, to assess the effectiveness of the manipulation of the intention of provocation, the attribution of intention measure was subjected to a 2 gender (men vs. women) x 2 type of provocation (hostilely-motivated vs. ambiguously-motivated) ANOVA with repeated measures on the latter variable. Second, to test for gender differences, the attribution of intention and retaliation was each subjected to a 2 Gender (men vs. women) x 2 type of provocation (hostilely motivated vs. ambiguously motivated) MANOVA with repeated measures on the latter variable followed by separate ANOVAs on each measure. Third, correlational analyses tested the hypothesized associations between the measures. Fourth, the Sobel test was used to test the hypothesis that the relation between trust beliefs in others and retaliation was mediated by the attribution of intention. The data were collapsed across the two types of motivation vignettes to yield a parsimonious and robust test of mediation.

Fifth, regression analyses (Cohen et al., 2003) examined the extent to which trust beliefs in others (total and by basis) were associated in a linear and quadratic fashion with the attribution of hostile intentions and retaliation. The analyses were carried out separately for two types of motivation (hostilely-motivated and ambiguously-motivated) to test whether those conformed to a cue-distortion or comprehensive personal style principles. The linear effects were statistically controlled in analyses of the quadratic relations as recommended (see Cohen et al., 2003). An initial set of analyses showed that gender did not appreciably moderate any of the observed relations.

6.4.2 | Manipulation check on the intention of provocation

The ANOVA yielded main effects of type of provocation, $F(1,137) = 1233.99$, $p < .001$, $\eta^2 = .90$ and of Gender, $F(1,137) = 4.78$, $p = .03$, $\eta^2 = .034$. The effectiveness of the manipulation was confirmed. The attribution of intention was greater to hostilely motivated provocation, $M = 34.80$, $SD = 0.43$ than to ambiguously motivated provocation, $M = 17.58$, $SD = 0.42$. The gender effect was the result of women ascribing greater attribution of intention to provocation, $M = 26.95$, $SD = 0.47$ than men, $M = 25.43$, $SD = 0.51$. The gender difference may have been due to women viewing acts of peer aggression as unexpected and consequently as hostility motivated.

6.4.3 | Sex differences in retaliation

The ANOVA on retaliation yielded the expected main effect of Type of Provocation, $F(1,137) = 526.51$, $p < .001$, $\eta^2 = .79$ (large effect size) which was qualified by a gender x type of provocation interaction, $F(1,137)=7.80$, $p = .006$, $\eta^2 = 0.05$ (approaching medium effect size). As expected, retaliation was greater to hostilely-motivated provocation, $M = 17.86$, $SD = 0.44$ than to ambiguously-motivated provocation, $M = 9.75$, $SD = 0.23$. The interaction comprised men reporting greater retaliation to hostilely motivated provocation, $M = 18.92$, $SD = 0.65$, than women,

M = 16.74, *SD* = 0.59. There was no appreciable gender difference in retaliation to ambiguously motivated provocation (*M* = 9.89, *SD* = 0.43, and *M* = 9.62, *SD* = 0.39 for men and women, respectively).

The current findings are consistent with the finding that males demonstrated greater direct aggression and retaliatory aggression than women during childhood and adolescence (Archer, 2019; Card et al., 2008). The current findings demonstrate though, that men showed greater retaliatory aggression than women and that gender difference is found when provocation is the result of hostile intentions. These findings are not consistent with meta-analyses which showed that a gender difference in aggression is lessened when provocation is more clearly motivated by hostile intentions (Bettencourt & Miller, 1996),

6.4.4 | Correlations between the measures

Consistent with the confirmatory factor analysis, the three trust belief basis subscales were intercorrelated and each subscale was correlated with the trust belief (total) scale (Table 1). As expected, the attribution of intention was correlated with retaliation for each type of motivation of provocation. Trust beliefs (total) scale, the reliability trust beliefs subscale, and honesty trust beliefs subscale were negatively correlated with the attribution of intention and retaliation for each type of provocation motivation with the following exceptions. Although the correlation between honesty trust beliefs subscale and the attribution of intention for ambiguously-motivated provocation was positive as expected it was nonsignificant. Also, the emotional trust beliefs subscale was negatively correlated with retaliation but only for ambiguously-motivated provocation. Because trust beliefs were associated with the attribution of intention and retaliation to both hostilely-motivated and ambiguously-motivated provocation, the findings supported the comprehensive personal style rather than a cue distortion process.

6.4.5 | Mediation

The Sobel test was significant, $z = 2.16$, $p = .031$ showing that the attribution of intention mediated the relation between trust beliefs (total) scale and retaliation. This finding, combined with the correlations, supported the trust-intention-retaliation hypothesis (Hypothesis 1). This pattern is consistent with findings that trust beliefs in others are negatively associated with aggression during childhood and adulthood (e.g., Copp et al., 2017; LaMotte et al., 2016). The current findings showed, however, that trust beliefs in others are related to specifically retaliatory aggression. Also, in support of the hypothesis by Malti et al. (2013), the relations are attributable to trust beliefs in others decreasing the likelihood of attributing provocation to hostile intentions.

6.4.6 | Regression analyses

These yielded several linear relations (commensurate with the observed correlations) but those were qualified by quadratic relations for the following. There was a quadratic relation between (1) the trust beliefs (total) and the attribution of intention to hostilely-motivated provocation (Figure 1); (2) the reliability of trust beliefs and the attribution of intention to hostilely-motivated provocation (Figure 2); (3) the emotional trust beliefs and the retaliation to hostilely-motivated provocation (Figure 3a), (4) the emotional trust beliefs and the attribution of intention to hostilely-motivated provocation (Figure 3b), and (5) the emotional trust beliefs and the retaliation to ambiguously-motivated provocation (Figure 3c). In support of the trust-retaliation quadratic hypothesis (Hypothesis 3),

TABLE 1 Correlations between the measures (with means and SDs)

Measure	Mean	SD	RelTB	HonTB	EmotTB	HMPI	HMPR	AMPI	APPR
Trust beliefs scale/subscales									
Total	78.52	10.28	0.79***	0.86***	0.76***	−0.28***	−0.21*	−0.35***	−0.21*
Reliability (RelTB)	27.13	3.86		0.59***	0.35***	−0.23**	−0.30***	−0.24**	−0.37***
Honesty (HonTB)	25.79	4.58			0.45***	−0.18*	−0.25**	−0.16	−0.27**
Emotional (EmotTB)	25.61	4.37				−0.11	−0.14	−0.11	−0.20*
Hostilely−motivated provocation									
Attribution of Intention (HMPI)	34.88	5.12					0.46***	0.35***	0.22**
Retaliation (HMPR)	17.75	5.23						0.38***	0.59***
Ambiguously-motivated provocation									
Attribution if Intention (AMPI)	17.64	4.95							0.63***
Retaliation (AMPR)	9.74	3.38							

Note: *Dfs* = 137.
*$p < .05$; **$p < .01$; ***$p < .001$.

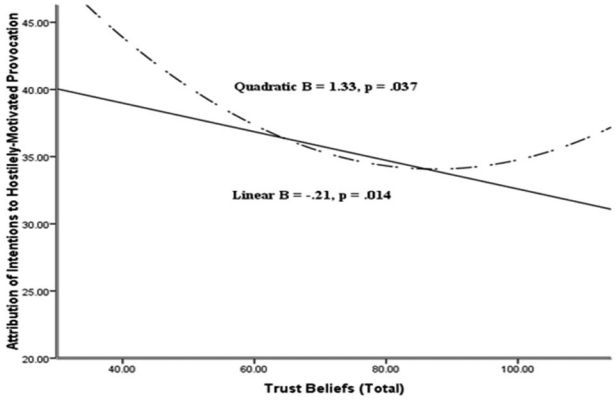

FIGURE 1 The linear and curvilinear relations between trust beliefs (total) and the attribution of intention to hostilely-motivated provocation

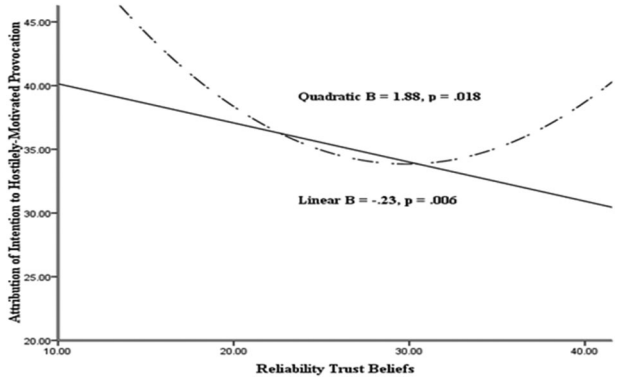

FIGURE 2 The linear and curvilinear relations between reliability trust beliefs and attribution of intentions to hostilely-motivated provocation

young adults with very low trust beliefs and those with very trust beliefs on the scales/subscale showed greater retaliation and attribution of intention than did those participants with the middle range of those trust beliefs. The findings complement the quadratic relations previously found between trust beliefs and retaliatory aggression in children and early adolescents (Rotenberg et al., 2013, 2014).

The findings also yielded support for the emotional betrayal hypothesis (Hypothesis 5). Three of the five observed quadratic relations were found with emotional trust beliefs. Those three quadratic relations were observed for the four possible combinations of the two measures (attribution of intention and retaliation) and the two types of motivation (ambiguously-motivated and hostilely-motivated). Only the quadratic relation with honesty trust beliefs was not predicted. The findings support the hypothesis that young adults with high emotional trust beliefs would view peer aggression as acts of betrayal (Hypothesis 5) and therefore view those as hostilely motivated and retaliate. It should be highlighted that this study provides evidence for young adults' perceptions of the intentions of the provocation and reports of how they would behave in those contexts. The next study (Study 2) examined the linear and curvilinear relations between trust beliefs in others and both anger and critical evaluation in the context of developing peer relationships (Hypothesis 4).

7 | STUDY 2

7.1 | Method

7.1.1 | Participants

The participants were 88 women (Mage = 21.5, *SD* = 5.3) who were undergraduates enrolled in their first year in a modest size university in the UK. They were an opportunity sample. They were solicited by posted advertisements and website advertisements on campus.

7.2 | Measures

7.2.1 | Trust beliefs

This was assessed by the same scale used in Study 1. The GTBS-LA total scale demonstrated acceptable internal consistency, $\alpha = .80$. The three trust belief subscales (reliability, honesty, and emotional) demonstrated acceptable internal consistency given the limited number of items per subscale, αs = .68, .60, and .60, respectively. Larger scores on the scale/subscales denoted greater trust beliefs.

7.2.2 | Anger

This was assessed by the UWIST mood adjective checklist which requires self-reports of 16 emotions on 5-point Likert scales (Matthews et al., 1990). Factor analyses of the UWIST have yielded an anger factor which was associated with a standardized measure of trait anger (Scott et al., 2001). The anger scale items included: anger, irritability, and hostility. The three-item scale demonstrated acceptable internal consistency, $\alpha = .68$. The items were summed to yield the scale. Larger numbers denoted greater anger.

7.2.3 | Critical evaluation

This was assessed by the participant's ratings of the qualities of the exchange between her and peer on 5-point Likert scales. The qualities were (1) cooperative (reverse coded), (2) unharmoniously, (3) friendly (reverse coded), (4) cold, and (5) unfocused. The 5-item critical evaluation scale demonstrated acceptable internal consistency, $\alpha = .71$. The items were summed to construct a scale. Larger numbers denoted a greater critical evaluation of a developing peer relationship.

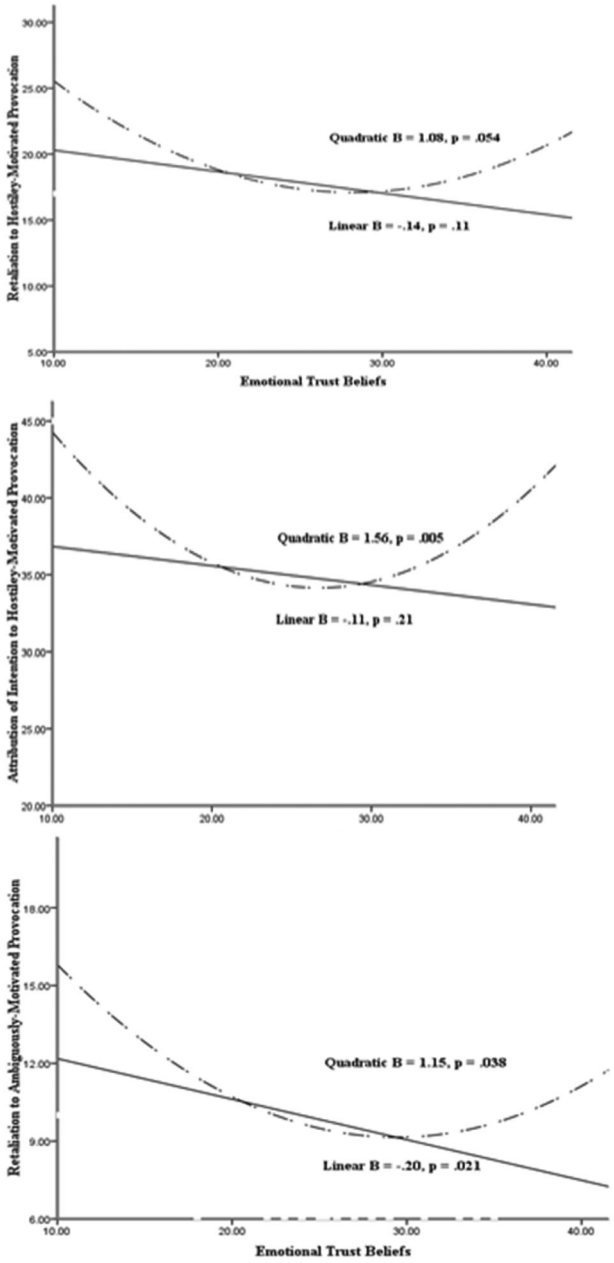

FIGURE 3 (a) The linear and curvilinear relations between emotional trust beliefs, (b) retaliation to hostilely-motivated provocation; attribution of intention to hostilely-motivated provocation; and (c) retaliation to ambiguously-motivated provocation

7.3 | Procedure

Participants were tested individually in a study described to them as one investigating "attitudes and relationship development". The participant was administered the GTBS-LA and the UWIST Mood Adjective Checklist. She then participated in a conversation with a same-sex student (a confederate of the experimenter) for the stated purpose of "getting acquainted with her." The participant selected a topic from a list that varied in intimacy (Miller et al., 1983) and then talked about it. The partner responded by choosing the likes/dislikes topic and then talking about it. This topic is of modest intimacy and appropriate for the early stage of relationship development (see Miller et al., 1983). Care was taken to ensure that the preset liking/disliking disclosure was spontaneous and natural. After the exchange, the experimenter stated that the study had to end then because of the shortness of time. The participant was asked to complete her ratings of the quality of the conversation. A brief conversation with pre-set disclosure procedure was employed to assess the extent to which the participant held a critical evaluation of the developing peer relationship (Hypothesis 2) that was not confounded with the varying disclosure of a peer over extended interactions.

7.4 | Results and discussion

7.4.1 | Strategies for the analyses

First, the data were subjected to correlational analyses to test the hypothesized associations. Second, the Sobel test was applied to test the hypothesis that anger mediated the relation between trust beliefs in others and critical evaluation (Hypothesis 1). Third, regression analyses were carried out to test for the predicted linear and curvilinear (i.e., quadratic) relations between trust beliefs and both anger and critical evaluation (Hypothesis 2).

7.4.2 | Correlations between the measures

The correlations (with *M*s and SDs) are shown in Table 2. Consistent with the SEM analyses in Study 1, there were intercorrelations between each of the trust belief subscales, and each subscale was correlated with trust beliefs (total) scale. As expected, anger was correlated with critical evaluation. In support of the trust-anger-critical evaluation hypothesis (Hypothesis 2), anger was negatively correlated with trust beliefs (total) and each of the three subscales (reliability, honesty, and emotional). As support for the trust-anger-critical evaluation hypothesis (Hypothesis 2), critical evaluation was negatively correlated with the trust beliefs (total) as well as the reliability trust subscale, and emotional trust subscale. Although the critical evaluation was negatively correlated with the honesty trust subscale, it was nonsignificant.

7.4.3 | Mediation

The Sobel test was significant ($z = -2.03$, $p = .043$) supporting the hypothesis that the relation between trust beliefs in others and critical evaluation is mediated by anger. The findings supported the trust-anger-critical evaluation hypothesis (Hypothesis 2). Young adults' trust beliefs in others were negatively associated with anger and critical evaluations of an emerging peer relationship. Anger mediated the relation between trust beliefs and critical evaluation.

TABLE 2 Correlations between the measures (with means and *SDs*)

Measure	Mean	SD	RelTB	HonTB	EmotTB	Anger	CE
Trust Beliefs Scales/Subscales							
Total	76.66	11.65	0.78***	0.84***	0.73***	−0.32**	−0.26*
Reliability (RelTB)	25.16	5.10		0.62***	0.38***	−0.25*	−0.27*
Honesty (HonTB)	24.49	5.13			0.46***	−0.27*	−0.17
Emotional (EmotTB)	25.63	4.68				−0.23*	−0.30**
Anger (Anger)	6.60	1.93					0.37**
Critical Evaluation (CE)	11.47	5.27					

Note: *Dfs* = 86.
*$p < .05$; **$p < .01$; ***$p < .001$.

7.4.4 | Regression analyses

The linear relations were qualified by the quadratic relations between (1) trust beliefs (total) and anger (Figure 4); (2) honesty trust beliefs and anger (Figure 5); (3) emotional trust beliefs and anger (Figure 6); (4) trust beliefs (total) and critical evaluation (Figure 7); and emotional trust beliefs and critical evaluation (Figure 8). The linear relations replicate the observed correlations. In support of both the trust-anger-critical quadratic hypothesis (Hypothesis 4) and emotional-betrayal hypothesis (Hypothesis 5), the quadratic curves showed that participants with very low emotional trust beliefs and those with very high emotional trust beliefs demonstrated greater anger and critical evaluation than did participants with the middle range of trust beliefs. In addition, quadratic relations were found between anger and both trust beliefs (total) and reliability trust beliefs. The former quadratic relation may be due, in part, to the fact that trust beliefs (total) scale included the emotional trust beliefs subscale items. The quadratic relation with reliability trust beliefs may be taken to suggest that some other facets of trust (expectation of fulfilling promises) contribute to quadratic relation with anger.

8 | GENERAL DISCUSSION

The findings from the current study yield support for both the TPPA hypothesis and the CAD. The current findings supported the CAT by quadratic relations between trust beliefs in others and interpersonal hostility. As expected, young adults with very high trust beliefs as well as very low trust beliefs showed greater attribution of intention for provocation, retaliatory aggression, anger, and critical evaluation of a developing peer relationship than those with the middle range of trust beliefs in others. The lower quadratic of the quadratic curve comprises young adults with very low trust beliefs who, according to the hypotheses, tend to be socially disengaged and show paranoid-like mistrust cognitions which dispose them to interpersonal hostility. The upper quadratic of the curve comprises young adults with very high trust beliefs who experience emotional harm in response to peer aggression, which disposes them to interpersonal hostility. As with the linear relations, the quadratic relations were obtained across different paradigms (vignette and relationship development) which attest to the breadth of the phenomenon.

The trust promotes psychosocial adjustment hypothesis has been advanced by Attachment Theory (Cohn, 1990; De Winter et al., 2016), Generalized Expectancy Theory (Rotter, 1980), and Social Capital Theory (Cozzolino, 2011). In support of the hypothesis in the current study, young adults' trust beliefs in others were negatively associated with interpersonal hostility as shown by lower levels of attributions of intention for provocation, retaliatory aggression, anger, and critical evaluation of a developing peer relationship. The linear relations were obtained across different paradigms (vignette and lab-based relationship development) which attest to the breadth of the phenomenon. Specifically, there were a number of linear relations that were not qualified by quadratic relations primarily, although not exclusively, regarding reliability and honesty bases of trust beliefs in others. The capacity to differentiate between the TPPA hypothesis and the CAD on the basis of trust beliefs is difficult because of the substantive correlations between the three bases of trust beliefs. Furthermore, the three bases of trust beliefs in others are subscales of the total trust beliefs scale and therefore the observed linear and quadratic relations with the total scales do not provide differential evidence for the two approaches.

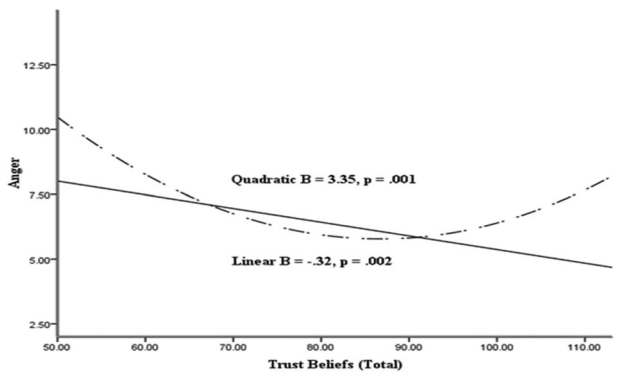

FIGURE 4 The linear and curvilinear relations between trust beliefs total and anger

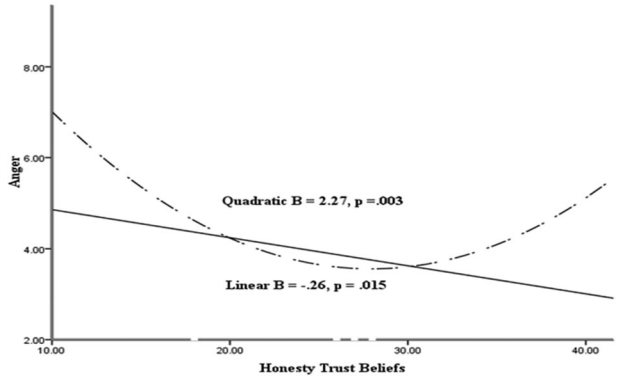

FIGURE 5 The linear and curvilinear relations between honesty trust beliefs and anger

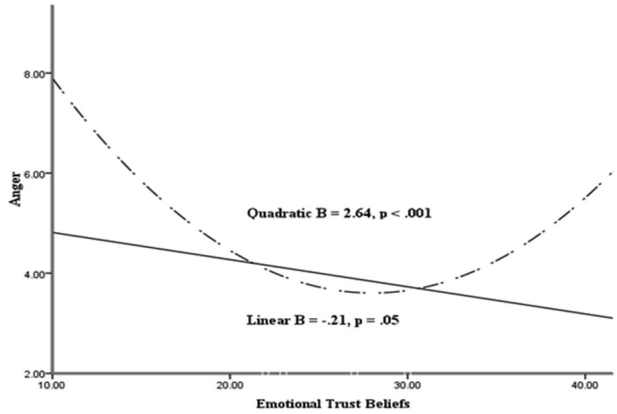

FIGURE 6 The linear and curvilinear relations between emotional trust beliefs and anger

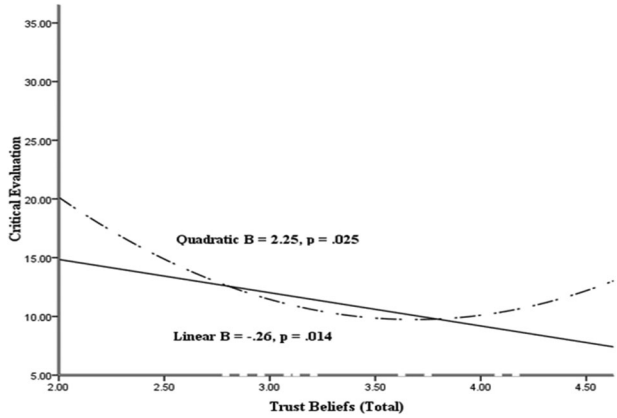

FIGURE 7 The linear and curvilinear relations between trust beliefs (total) and critical evaluation

8.1 | Limitations and directions for future research

In the future, longitudinal investigations are needed to identify probable causal relations between trust beliefs in others and interpersonal hostility in young adults. Longitudinal studies that track individuals from childhood to young adulthood also would be useful to examine the stability of linear and quadratic relations across the course of development. The current research examined young adults' reports of their attribution of hostile intentions and retaliatory aggressive behavior for hypothetical vignette scenarios. In the future, it would be worthwhile to examine whether the corresponding relations would be found in young adults' actual social interactions with peers. In a related vein, anger could be assessed by physiological measures as well as self-reports of affect in future research. Study 2 showed that young women with extreme trust beliefs in others dispose them to hold critical views in the anticipation and development of a peer relationship. Those dispositions could undermine their development of longer-term peer relationships.

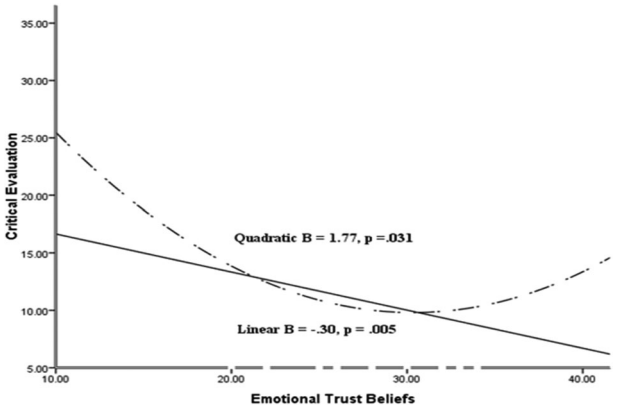

FIGURE 8 The linear and curvilinear relations between emotional trust beliefs and critical evaluation

The research was guided by the principle that women with extreme trust beliefs had a social history that disposed them to experience anger toward others and critical evaluations of peer relationships. It has been found that girls with very high trust beliefs and those with very low beliefs engaged in greater indirect aggression, elicited lower group affiliation, incurred greater peer rejection, and showed greater distress in peer interactions than did girls with the middle range of trust beliefs (Rotenberg et al., 2014). The current study confirmed the expected implications of gender-based social histories regarding the relation between trust beliefs in others and interpersonal hostility as evidenced by anger towards others and critical evaluation of a developing peer relationship. Study 2 did not examine young adult men because the research had not shown that they had a similar social history. Also, the relationship development method employed in the study was suited for assessing women, particularly in universities. A different research paradigm might be employed for men such as their aggressive exchanges in a competitive computer game (see Archer, 2019).

The current study examined the relationship between trust beliefs in others and their dispositions to interpersonal hostility. A body of research has shown that aggression in social interactions has dyadic properties that bear on interpersonal hostility. Anderson et al. (2008) found that young adults' aggressive dispositions combine in a dyadic fashion to establish a hostile environment. The findings demonstrate that aggressive dispositions of partners in dyadic exchanges contribute to an

escalation of aggression. The dyadic nature of aggressive interactions is shown by actor-partner interdependence and relationship effects in aggressive exchanges by social relation model analyses (Coie et al., 1999; O'Leary & Slep, 2003). In the context of the current findings, young adults with low-levels and very low levels of trust beliefs are prone to attribute hostile intentions to provocation, engage in retaliatory aggression, experience heightened anger towards others (for women), and experience heightened critical evaluation of a developing peer relationship (for women). Guided by Anderson et al. (2008) hostile environment formulation, those dispositions could cause the young adults to establish a hostile environment in dyadic interactions which involve an escalation of aggression. A similar process would apply to young adults who hold very high trust beliefs in others as it bears on emotional trust. Once those individuals experience betrayal of trust early in the interaction, their interpersonal hostility disposition would result in establishing a hostile environment and corresponding escalation of aggression.

Trust beliefs in others have dyadic properties as well. Individuals in dyadic interactions are prone to match each other's trust beliefs and trusting behaviors (Rotenberg, 2010). Young adults show dyadic patterns of trust and, when betrayed, show escalations of distrust in romantic relationships (Uysal et al., 2012). Social relations model analyses have shown that trust in social relationships has dyadic properties (see Betts et al., 2014). The dyadic properties of aggression and trust beliefs may be synchronized within dyadic interactions. After an act of aggression occurs, aggressive exchanges and distrust beliefs would concurrently escalate during subsequent dyadic exchanges. This escalation would occur optimally when both individuals in the dyad had the dispositions for aggression and for extreme trust beliefs in others. These hypotheses warrant further investigation.

8.2 | Summary

The research showed that young adults' trust beliefs in others were associated with interpersonal hostility in a linear and a curvilinear fashion. Young adults' trust beliefs in others were linearly related to multiple measures of interpersonal hostility. The research showed, though, that there were quadratic relations between young adults' trust beliefs in others and interpersonal hostility. Young adults with extreme trust beliefs in others, primarily emotional based, showed greater interpersonal hostility than young adults with the middle range of trust beliefs. The findings supported both the trust promotes psychosocial adjustment hypothesis and the centralist approach to trust.

CONFLICT OF INTERESTS

There are no conflict of interests to declare.

PEER REVIEW

The peer review history for this article is available at https://publons.com/publon/10.1002/ab.21969

DATA AVAILABILITY STATEMENT

The data for Study 1 is available by the OSF by the link: https://osf.io/gdnq5/?view_only=f43ee4063698469b9f1ed2e352905a03

The data for Study 2 is available by the OSF by the link: https://osf.io/gdnq5/?view_only=f43ee4063698469b9f1ed2e352905a03

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